

Kantorovich potential φ (for the cost $c(x, y) = |x - y|^2$) in the transport from ρ_0 to ν is unique up to additive constants, then the first variation of F at ρ_0 is given by $\varphi + V$, and we can apply Proposition 7.20 to obtain the existence of a constant C such that

$$\varphi + V \geq C \text{ on } \Omega, \quad \varphi + V = C \text{ on } \text{spt}(\rho_0).$$

This is the case if $\text{spt}(\nu) = \Omega$, for instance.

Another way to obtain the optimality conditions is the following: the second term of F is a continuous and linear functional on $\mathcal{M}(\Omega)$, and the first is a convex functional defined on $\mathcal{M}(\Omega)$, with subdifferential given by the set of Kantorovich potentials. Hence, we can prove

$$\partial F(\rho_0) = \{V + \varphi : \varphi \text{ Kantorovich potential from } \rho_0 \text{ to } \nu\}.$$

From standard optimality conditions in convex analysis we obtain the existence of a potential φ such that $V + \varphi = 0$. In this case the result that we get is much stronger with this convex procedure, since we get rid of the inequality case. Note, however, that the function φ that we get is not necessarily c -concave: since we know $\varphi = \varphi^{cc}$ on $\text{spt}(\rho_0)$, this means that it has been modified out of $\text{spt}(\rho_0)$ in order to guarantee the equality.

Example 7.22. Consider

$$\min \left\{ F(\rho) := W_2^2(\rho, \nu) + \int_{\Omega} \frac{1}{p} \rho^p(x) dx : \rho \in \mathcal{P}(\Omega), \rho \ll \mathcal{L}^d \right\},$$

where $\Omega \subset \mathbb{R}^d$ is compact and $p > 1$. Again, a minimizer $\rho_0 \in L^p(\Omega)$ exists and is regular (in the sense of Definition 7.12), since F is finite on $L^p(\Omega) \cap \mathcal{P}(\Omega)$ which includes for sure all convex combinations of ρ_0 and L^∞ functions. Computing the first variation and using Proposition 7.20, we obtain the existence of a constant C such that

$$\varphi + \rho^{p-1} \geq C \text{ on } \Omega, \quad \varphi + \rho^{p-1} = C \text{ on } \text{spt}(\rho_0).$$

This, by the way, can be re-written as $\rho^{p-1} = (C - \varphi)_+$.

On the other hand, even if the functional is still convex, it is not easy to compute the subdifferential of F , viewed as a functional on $\mathcal{M}(\Omega)$ (set to $+\infty$ outside L^p probabilities). Indeed, the subdifferential of a sum is not always the sum of the subdifferentials (see for instance [158] for convex analysis issues in infinite dimension; however, this requires continuity of at least one of the two functionals, while here both are finite on a small subset of $\mathcal{M}(\Omega)$). Moreover the first variation of the second term, i.e. ρ^{p-1} , does not belong in general to the space $C(\Omega)$, i.e. to the space $\mathcal{M}(\Omega)$ is in duality with. A way of circumventing this difficulty would be to view F as a functional on $L^p(\Omega)$. Here, the second term becomes a continuous convex functional. The first one must be re-written as $\sup\{\int \phi d\rho + \int \phi^c d\nu : \phi \in L^{p'}(\Omega)\}$ (with a proper definition of the c -transform in terms of essential infima instead of infima). This would provide the optimality condition: there exists an $L^{p'}$ Kantorovich potential φ such that $\varphi + \rho^{p-1} = 0$. Again, $\varphi = \varphi^{cc}$ ρ_0 -a.e., and $\varphi \leq \varphi^{cc}$ elsewhere. This gives again

$$\varphi^{cc} + \rho^{p-1} \geq C \text{ on } \Omega, \quad \varphi^{cc} + \rho^{p-1} = C \text{ on } \text{spt}(\rho_0).$$

7.3 Displacement convexity

In all the considerations of the previous section about convexity, we always considered the standard convex interpolations $[0, 1] \ni t \mapsto (1-t)\mu_0 + t\mu_1$. Yet, another notion of convexity, more linked to the metric structure of the space $\mathbb{W}_p(\Omega)$, may be useful.

Let us start from this general definition.

Definition 7.23. In a geodesic metric space X , we define $F : X \rightarrow \mathbb{R} \cup \{+\infty\}$ to be *geodesically convex* if for every two points $x_0, x_1 \in X$ there exists a constant speed geodesic ω connecting $\omega(0) = x_0$ to $\omega(1) = x_1$ such that $[0, 1] \ni t \mapsto F(\omega(t))$ is convex.

This obviously reduces to usual convexity in $\mathbb{R}^d$ or in any other normed vector space where segments are the unique geodesics (note that in any normed vector space segments are geodesics but they may not be the unique ones, as it is the case in L^1 or L^∞ or any non-strictly convex space). By the way, in spaces where there is not uniqueness of the geodesics the definition we gave could be discussed, since one could also choose to define as geodesically convex those functionals that satisfy the geodesic inequality on *every* geodesic ω . . . But the one we gave is the definition that is usually chosen, as it satisfies some extra stability results that would not be true with the more restrictive one requiring convexity for every geodesic.

The notion of geodesic convexity in the space $\mathbb{W}_p(\Omega)$ has been introduced by McCann in [229], and is particularly interesting since we know how to characterize geodesics in such a space. This notion of convexity is usually referred to as *displacement convexity*. Note that it could a priori depend on the exponent p .

7.3.1 Displacement convexity of $\mathcal{V}$ and $\mathcal{W}$

We recall that the geodesics for the W_p distance are of the form $\mu_t = (\pi_t)_\# \gamma$ where γ is an optimal transport for $c(x, y) = |x - y|^p$ and $\pi_t(x, y) = (1-t)x + ty$.

Proposition 7.24. *The functional $\mathcal{V}$ is displacement convex if and only if V is convex. The functional $\mathcal{W}$ is displacement convex if W is convex.*

Proof. First consider $\mathcal{V}$ and suppose that V is convex. Let us evaluate $\mathcal{V}(\mu_t)$:

$$\mathcal{V}(\mu_t) = \int V d(\pi_t)_\# \gamma = \int V((1-t)x + ty) d\gamma.$$

It is clear from this formula that $t \mapsto \mathcal{V}(\mu_t)$ is convex if V is convex.

On the other hand, the convexity of V is a necessary condition for $\mathcal{V}$ being convex as one can easily check by considering geodesics of the form $\mu_t = \delta_{(1-t)x + ty}$ since $\mathcal{V}(\mu_t) = V((1-t)x + ty)$.

The proof for the convexity of $\mathcal{W}$ is similar: consider

$$\begin{aligned} \mathcal{W}(\mu_t) &= \int W(x, x') d(\pi_t)_\# \gamma(x) d(\pi_t)_\# \gamma(x') \\ &= \int W((1-t)x + ty, (1-t)x' + ty') d\gamma(x, y) d\gamma(x', y'), \end{aligned}$$

which easily gives the condition for the convexity. $\square$ $\square$

Note that we did not state that convexity of W is a necessary condition for $\mathcal{W}$, since it is not true in general. Consider for instance the following 1D case.

Proposition 7.25. *Let $\Omega = (a, b)$, $\Delta^+ = \{(x, y) \in \Omega \times \Omega, y \geq x\}$ and $\Delta^- = \{(x, y) \in \Omega \times \Omega, y \leq x\}$. Then it is sufficient that W is convex when restricted to Δ^+ and Δ^- , in order to have displacement convexity of $\mathcal{W}$.*

Proof. Just consider the proof of Proposition 7.24 and check that all the segments $t \mapsto ((1-t)x + ty, (1-t)x' + ty')$ for $(x, y), (x', y') \in \text{spt}(\gamma)$ are contained either in Δ^+ or in Δ^- . This is true thanks to the monotonicity properties of $\text{spt}(\gamma)$ that we showed in Section 2.2. Actually, we showed that, for every strictly convex cost, the (unique) optimal transport plan γ satisfies

$$(x, y), (x', y') \in \text{spt}(\gamma), \quad x < x' \quad \Rightarrow \quad y \leq y'.$$

This means that if $x < x'$ then we have $(x, x') \in \Delta^+$ and $(y, y') \in \Delta^+$ and, by convexity, $((1-t)x + ty, (1-t)x' + ty') \in \Delta^+$. Analogously, for $x > x'$ we get that the segment is contained in Δ^- . If $x = x'$ it is enough to look at the transport plan from the point of view of y , thus getting the implication $y < y' \Rightarrow x \leq x'$ and concluding in the same way. The only case that stays apart is $x = x'$ and $y = y'$ but in this case the segment reduces to a point.

This proves convexity of $\mathcal{W}$ along the geodesics in $\mathbb{W}_p$ for $p > 1$. It also works for W_1 if we choose the same γ as an optimal transport plan (in this case it will not be the unique one, but this will be enough to build a geodesic where there is convexity). $\square$

An interesting consequence of this criterion is the fact that the squared dual norm $\mu \mapsto \|\mu\|_{\mathcal{X}}^2$ for $\mathcal{X} = H^1([-1, 1])$ is actually displacement convex, as a corollary of the characterization of Proposition 7.6 (but it does not work for $\mathcal{X} = H_0^1$). This has been pointed out and used in [111].

This is an example of displacement convex functional which involves, somehow, derivatives. We could say that, as an H^{-1} norm, it is a functional of order -1 . Displacement convex functionals of order different than 0 are not so common to find. For the first-order case, the only known example is contained in [116] and it is, again, only available in 1D. Nothing is known in the multidimensional case.

7.3.2 Displacement convexity of $\mathcal{F}$

The most interesting displacement convexity result is the one for functionals depending on the density.

To consider these functionals, we need some technical facts.

The starting point is the computation of the density of an image measure, via standard change-of-variable techniques, as we saw in the Memo Box 1.14.

Then, we underline an interesting computation:

Lemma 7.26. *Let A be a $d \times d$ matrix such that its eigenvalues λ_i are all real and larger than -1 (for instance this is the case when A is symmetric and $I + A \geq 0$). Then $[0, 1] \ni t \mapsto g(t) := \det(I + tA)^{1/d}$ is concave.*

Proof. We can write A in a suitable basis so that it is triangular, and we get

$$g(t)^d = \prod_{i=1}^d (1 + t\lambda_i).$$

Differentiating,

$$dg(t)^{d-1} g'(t) = \sum_{j=1}^d \lambda_j \prod_{i=1, i \neq j}^d (1 + t\lambda_i) = g(t)^d \sum_{j=1}^d \frac{\lambda_j}{1 + t\lambda_j},$$

i.e. $dg'(t) = g(t) \sum_{j=1}^d \frac{\lambda_j}{1 + t\lambda_j}$. If we differentiate once more we get

$$dg''(t) = g'(t) \sum_{j=1}^d \frac{\lambda_j}{1 + t\lambda_j} - g(t) \sum_{j=1}^d \frac{\lambda_j^2}{(1 + t\lambda_j)^2}.$$

Here we use the quadratic-arithmetic mean inequality which gives

$$\sum_{j=1}^d \frac{\lambda_j^2}{(1 + t\lambda_j)^2} \geq \frac{1}{d} \left(\sum_{j=1}^d \frac{\lambda_j}{1 + t\lambda_j} \right)^2 = d \left(\frac{g'(t)}{g(t)} \right)^2$$

and hence

$$dg''(t) \leq d \frac{g'(t)^2}{g(t)} - d \frac{g'(t)^2}{g(t)} = 0,$$

which proves the concavity of g . $\square$

Remark 7.27. We observe that in dimension $d = 1$ the above Lemma is straightforward, and that g is actually affine. $\square$

We can now state the main theorem.

Theorem 7.28. *Suppose that f is convex and superlinear, $f(0) = 0$ and that $s \mapsto s^{-d} f(s^d)$ is convex and decreasing. Suppose that Ω is convex and take $1 < p < \infty$. Then $\mathcal{F}$ is geodesically convex in $\mathbb{W}_p$.*

Proof. Let us consider two measures μ_0, μ_1 with $\mathcal{F}(\mu_0), \mathcal{F}(\mu_1) < +\infty$. They are absolutely continuous and hence there is a unique constant speed geodesic μ_t between them (see Theorem 5.27 and Proposition 5.32), which has the form $\mu_t = (T_t)_\# \mu_0$, where $T_t = \text{id} + t(T - \text{id})$. Note that T_t is injective because of Lemma 4.23.

We first look at the case $p = 2$, which is easier. In this case we have $T_t(x) = x - t\nabla\varphi(x)$, where φ is such that $\frac{x^2}{2} - \varphi$ is convex. This implies, by Theorem 3.16, that $\nabla\varphi$ is countably Lipschitz, and so is T_t . Hence they are approximately differentiable a.e.³. The Hessian $D^2\varphi$ (or, equivalently, the approximate gradient of $\nabla\varphi$), is symmetric and $D^2\varphi \leq I$. Let us define $A(x) = -D^2\varphi(x)$. From the formula for the density of the

³Actually, it is also known that $D^2\varphi$ exists a.e.: indeed, convex functions are twice differentiable a.e. (see for instance [161], the original result being stated in [6]).

image measure, we know that μ_t is absolutely continuous and we can write its density ρ_t as $\rho_t(y) = \rho(T_t^{-1}(y)) / \det(I + tA(T_t^{-1}(y)))$ and hence

$$\mathcal{F}(\mu_t) = \int f\left(\frac{\rho(T_t^{-1}(y))}{\det(I + tA(T_t^{-1}(y)))}\right) dy = \int f\left(\frac{\rho(x)}{\det(I + tA(x))}\right) \det(I + tA(x)) dx,$$

where we used the change of variable $x = T_t^{-1}(y)$, which gives $y = T_t(x)$ and $dy = \det DT_t(x) dx = \det(I + tA(x)) dx$.

From Lemma 7.26 we know that $\det(I + tA(x)) = g(t, x)^d$ for a function $g : [0, 1] \times \Omega$ which is concave in t . It is a general fact that the composition of a convex and decreasing function with a concave one gives a convex function. This implies that

$$t \mapsto f\left(\frac{\rho(x)}{g(t, x)^d}\right) g(t, x)^d$$

is convex (if $\rho(x) \neq 0$ this uses the assumption on f and the fact that $t \mapsto g(t, x)/\rho(x)^{\frac{1}{d}}$ is concave; if $\rho(x) = 0$ then this function is simply zero). Finally, we proved convexity of $t \mapsto \mathcal{F}(\mu_t)$.

We now have to adapt to the case $p \neq 2$. In this case, setting $h(z) = \frac{1}{p}|z|^p$, we have $T_t(x) = x - t\nabla h^*(\nabla\varphi(x))$. Note that both h and h^* are $C^2(\mathbb{R}^d \setminus \{0\})$. First, we note that we can decompose, up to negligible sets, Ω into two measurable parts: the set Ω' where $\nabla\varphi = 0$ and the set Ω'' where $\nabla\varphi \neq 0$. The analysis that we do is very much similar to what we did in Section 3.3.2. Ω' is the set where T is the identity, and so is T_t . Since T_t is injective, the density is preserved on this set, and we can apply Lemma 7.26 with $A = 0$. The set Ω'' , on the contrary, can be decomposed into a countable union of sets Ω_{ij} where $x \in B_i$, $T(x) \in B_j$, $(B_i)_i$ is a countable family of balls generating the topology of $\mathbb{R}^d$, and we only take pairs such that $B_i \cap B_j = \emptyset$. On this sets $\varphi = \psi^c$ also coincides with φ_{ij} defined as the restriction to B_i of the function $x \mapsto \inf_{y \in B_j} h(x - y) - \psi(y)$, which is λ -concave for $\lambda = \sup\{\|D^2h(z)\| : z \in B_j - B_i\}$. This proves that φ_{ij} has the same regularity of concave functions, and that $\nabla\varphi$ and $\nabla h^*(\nabla\varphi)$ are countably Lipschitz on Ω'' . In particular, $D^2\varphi$ exists a.e.

If we fix a point $x_0 \in \Omega''$ where $\nabla\varphi(x_0)$ and $D^2\varphi(x_0)$ exist, then we can write $\varphi(x) \leq h(x - T(x_0)) - \psi(T(x_0))$, an inequality which is true for every x , with equality at $x = x_0$. In particular we get $D^2\varphi(x_0) \leq D^2h(x_0 - T(x_0)) = D^2h(\nabla h^*(\nabla\varphi(x_0)))$. From general properties of Legendre transforms, we have

$$D^2h(\nabla h^*(z)) = [D^2h^*(z)]^{-1}$$

(just differentiate the relation $\nabla h(\nabla h^*(z)) = z$). Hence, we can apply Lemma 7.26 with $A(x) = -D^2h^*(\nabla\varphi(x))D^2\varphi(x)$, which is diagonalizable and has eigenvalues larger than -1 (see Box 7.2 below). $\square$ $\square$

Box 7.2. – Memo – Diagonalizing products of symmetric matrices

It is well known that symmetric matrices can be diagonalized on $\mathbb{R}$. A trickier result is the following:

Theorem - If A, B are symmetric $d \times d$ matrices, and A is positive-definite, then AB is diagonalizable.

We do not prove it, but we just prove the following weaker result.

Proposition - If A, B are symmetric $d \times d$ matrices, and A is positive definite, then AB has real eigenvalues.

Proof - suppose that $\lambda \in \mathbb{C}$ is an eigenvalue of AB , i.e. $AB\mathbf{v} = \lambda\mathbf{v}$ for a vector $\mathbf{v} \in \mathbb{C}^n \setminus \{0\}$. Take the Hermitian product with $A^{-1}\mathbf{v}$, i.e. $\bar{\mathbf{v}}^t B\mathbf{v} = \lambda \bar{\mathbf{v}}^t A^{-1}\mathbf{v}$, where the symbol t denotes transposition in the sense of matrices. Note that, by diagonalizing independently B and A^{-1} , both the terms $\bar{\mathbf{v}}^t B\mathbf{v}$ and $\bar{\mathbf{v}}^t A^{-1}\mathbf{v}$ are real, and $\bar{\mathbf{v}}^t A^{-1}\mathbf{v} > 0$. Hence, $\lambda \in \mathbb{R}$.

Another important point that we need is the following.

Proposition - If $B \leq A^{-1}$ (in the sense of symmetric matrices, i.e. $\mathbf{v}^t (A^{-1} - B)\mathbf{v} \geq 0$ for all $\mathbf{v}$), then all the eigenvalues λ_i of AB satisfy $\lambda_i \leq 1$

Proof - again, write $AB\mathbf{v} = \lambda\mathbf{v}$ and deduce $\bar{\mathbf{v}}^t B\mathbf{v} = \lambda \bar{\mathbf{v}}^t A^{-1}\mathbf{v}$. This implies $\bar{\mathbf{v}}^t A^{-1}\mathbf{v} \geq \bar{\mathbf{v}}^t B\mathbf{v} = \lambda \bar{\mathbf{v}}^t A^{-1}\mathbf{v}$, which implies $\lambda \leq 1$.

Let us see some easy example of convex functions satisfying the assumptions of Theorem 7.28. For instance

- for any $q > 1$, the function $f(t) = t^q$ satisfies these assumptions, since $s^d f(s^{-d}) = s^{-d(q-1)}$ is convex and decreasing;
- the entropy function $f(t) = t \log t$ also satisfies the assumptions since $s^d f(s^{-d}) = -d \log s$ is convex and decreasing;
- if $1 - \frac{1}{d} \leq m < 1$ the function $f(t) = -t^m$ is convex, and if we compute $s^d f(s^{-d}) = -t^{m(1-d)}$ we get a convex and decreasing function since $m(1-d) < 1$; yet these functions lack the superlinearity assumption, but this does not prevent us from applying the same proof of Theorem 7.28 to the case where μ_0 and μ_1 are supposed to be a priori absolutely continuous.

Let us see some consequences of Theorem 7.28 in the case $f(t) = t^q, q > 1$.

Proposition 7.29. Consider an exponent $1 < q \leq +\infty$ and two probability measures $\mu_0, \mu_1 \in L^q(\Omega)$ (in the sense that they are absolutely continuous and their densities are L^q). Take the (unique) geodesic μ_t connecting them in $\mathbb{W}_p$ (for $p > 1$). Then the measures μ_t are also L^q and $\|\mu_t\|_{L^q} \leq \max\{\|\mu_0\|_{L^q}, \|\mu_1\|_{L^q}\}$. When $q < +\infty$, we also have $\|\mu_t\|_{L^q}^q \leq (1-t)\|\mu_0\|_{L^q}^q + t\|\mu_1\|_{L^q}^q$.

Proof. The case $q < +\infty$ is an easy consequence of Theorem 7.28. Actually, if we use $f(t) = t^q$ we have $\mathcal{F}(\mu) = \|\mu\|_{L^q}^q$. Hence

$$\|\mu_t\|_{L^q}^q \leq (1-t)\|\mu_0\|_{L^q}^q + t\|\mu_1\|_{L^q}^q \leq (\max\{\|\mu_0\|_{L^q}, \|\mu_1\|_{L^q}\})^q.$$

This allows to get the desired L^q estimate.

The case $q = +\infty$ is just obtained by taking the q -th root and passing to the limit $q \rightarrow +\infty$. $\square$ $\square$

We finish this section observing that in 1D we can say something more.

Proposition 7.30. *Suppose $d = 1$, $\Omega = [a, b] \subset \mathbb{R}$ and take $q > 0$. Then the functional F defined through $F(\rho) = \int_a^b \rho(x)^{-q} dx$ for $\rho \ll \mathcal{L}^1$ is displacement convex in $\mathbb{W}_p(\Omega)$. In particular, if $\rho_0, \rho_1 \geq c > 0$, then we also have $\rho_t(x) \geq c$ for every $t \in [0, 1]$, where ρ_t is the geodesic connecting ρ_0 to ρ_1 .*

Proof. Notice the the function $f(s) = s^{-q}$ is convex for $s > 0$ and that $s \mapsto s^d f(s^{-d}) = s^{q+1}$ is convex but increasing. Yet, if we re-read the proof of Theorem 7.28 we observe that we only needed it to be decreasing to ensure that the composition with g was convex, but in this case g is affine, and convexity of $s \mapsto s^d f(s^{-d})$ is enough. Moreover, if we assume $F(\rho) < +\infty$ then we have $\rho > 0$ a.e., which shows that it is not necessary to assume $f(0) = 0$. This proves the desired displacement convexity. To get the lower bound on geodesics, we just need to act as in proposition 7.29, letting $q \rightarrow \infty$. $\square$ $\square$

7.3.3 Convexity on generalized geodesics

It is quite disappointing to note that the functional $\mu \mapsto W_2^2(\mu, \nu)$ is not, in general, displacement convex. This seems contrary to the intuition because usually squared distances are nice convex functions. However, we can see that this fails from the following easy example. Take $\nu = \frac{1}{2}\delta_{(1,0)} + \frac{1}{2}\delta_{(-1,0)}$ and $\mu_t = \frac{1}{2}\delta_{(t,a)} + \frac{1}{2}\delta_{(-t,-a)}$. The curve μ_t , if $a > 1$ is the geodesic between μ_{-1} and μ_1 (because the optimal transport between these measures sends $(a, -1)$ to $(a, 1)$ and $(-a, 1)$ to $(-a, -1)$). Yet, if we compute $W_2^2(\mu_t, \nu)$ we have

$$W_2^2(\mu_t, \nu) = a^2 + (1-t)^2 \wedge (1+t)^2.$$

But this function is not convex! (see Figure 7.1)

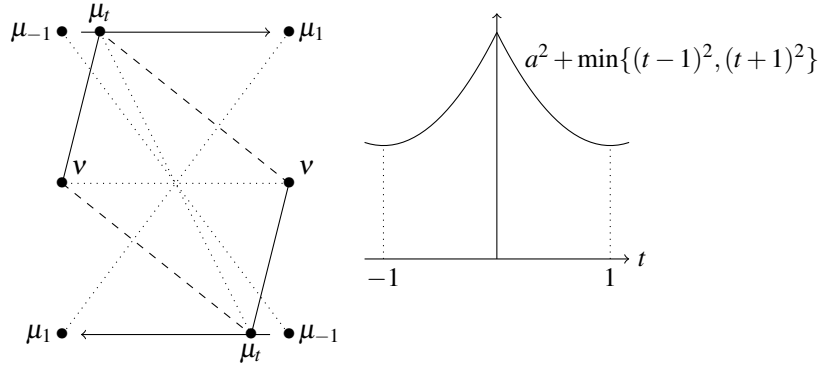


Figure 7.1: The distance $W_2^2(\mu_t, \nu)$.

The lack of geodesic convexity of this easy functional⁴ is a problem for many issues (in particular for some metric approaches to gradient flows through the EVI condition,

⁴By the way, this functional can even be prove to be somehow geodetically concave, as it is shown in [17], Theorem 7.3.2.

see Section 8.4.1) and an alternate notion has been proposed, namely that of convexity along generalized geodesics.

Definition 7.31. If we fix an absolutely continuous probability $\rho \in \mathcal{P}(\Omega)$, for every pair $\mu_0, \mu_1 \in \mathcal{P}(\Omega)$ we call *generalized geodesic* between μ_0 and μ_1 with base ρ in $\mathbb{W}_2(\Omega)$ the curve $\mu_t = ((1-t)T_0 + tT_1)_\# \rho$, where T_0 is the optimal transport map (for the cost $|x-y|^2$) from ρ to μ_0 , and T_1 from ρ to μ_1 .

It is clear that $t \mapsto W_2^2(\mu_t, \rho)$ satisfies

$$\begin{aligned} W_2^2(\mu_t, \rho) &\leq \int |((1-t)T_0(x) + tT_1(x)) - x|^2 d\rho(x) \\ &\leq (1-t) \int |T_0(x) - x|^2 d\rho(x) + t \int |T_1(x) - x|^2 d\rho(x) \\ &= (1-t)W_2^2(\mu_0, \rho) + tW_2^2(\mu_1, \rho) \end{aligned}$$

and hence we have the desired convexity along this curve. Moreover, similar considerations to those we developed in this section show that all the functionals that we proved to be geodesically convex are also convex along generalized geodesics. We do not develop these proofs here, and we refer to [17] for more details (for the case of the functional $\mathcal{F}$, Lemma 7.26 has to be changed into “ $t \mapsto \det((1-t)A + tB)^{1/d}$ is concave, whenever A and B are symmetric and positive-definite” (the proof is similar).

7.4 Discussion

7.4.1 A case study: $\min F(\rho) + W_2^2(\rho, \nu)$

Among variational problems involving optimal transportation and Wasserstein distances, a very recurrent one is the following

$$\min_{\rho \in \mathcal{P}_2(\Omega)} \frac{1}{2} W_2^2(\rho, \nu) + \tau F(\rho), \quad (7.2)$$

where F is a given functional on probability measures, $\tau > 0$ a parameter which can possibly be small, and ν is a given probability in $\mathcal{P}_2(\Omega)$ (the space of probability measures on Ω with finite second moment $\int |x|^2 d\rho(x) < +\infty$). This very instance of the problem is exactly the one we face in the time-discretization of the gradient flow of F in $\mathbb{W}_2(\Omega)$ (see Section 8.2).

But the same problem also appears, for fixed $\tau > 0$, in other frameworks as well. For instance in image processing, if F is a smoothing functional, this is a model to find a better (smoother) image ρ which is not so far from the original density ν (and the choice of the distance W_2 is justified by robustness arguments), see [209, 93]. In urban planning (see [94, 270]) ν can represent the distribution of some resources and ρ that of population, who wants to be close to ν but also to guarantee enough space to each individual. In this case the functional F favors diffuse measures (for instance using the functional $\mathcal{F}(\rho) = \int f(\rho(x)) dx$ for a convex and superlinear function f